

Login:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
Valid Username/Password	The Client and Server are both running concurrently.	User enters a valid Username - NicoBarr and valid password - strongMAN	User enters valid username and password Client sends login credentials and is a success System grants access to user	User logged in with valid username and password.	Ensure the user logged in with a valid username and password.
Invalid Password	The Client and Server are both running concurrently.	User enters a valid username - NicoBarr and invalid password - strongMA	User enters valid username and invalid password Client sends login credentials and is a failed login System does not grant access to the user	User is unable to log in due to invalid password.	Ensure the user is unable to log in with an invalid password even though the username is correct.
Invalid Username	The Client and Server are both running concurrently.	User enters a invalid Username - NicoBarr* and valid password - strongMAN	User enters invalid username and valid password Client sends login credentials and is a failed login System does not grant access to the user	User is unable to log in due to invalid username.	Ensure the user is unable to log in with an invalid username even though the password is correct.
Account lockout	The Client and Server are both running concurrently.	User enters a valid email - NicoBarr and invalid password -	User enters a valid email and invalid	User is unable to log in due to invalid password after 3 tries	Ensure the user is unable to log in with an invalid after 3 tries.

		strongMA for a 3rd time	password for a 3rd time Client sends login credentials to system and it is a fail System increments error count to 3 System does not grant access to user		
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Connect:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
Valid IP Address	The Client and Server are both running concurrently.	User enters a valid IP Address; 127.0.0.1	Server send the IP Address to Client Client verifies IP address is valid System connects to Server with the IP address	System connection is successful	Ensure IP address is valid and there is successful connection
Invalid IP Address	The Client and Server are both running concurrently.	User enters a invalid IP Address; 127.0.0.&	Server send the IP Address to Client Client verifies IP address is valid - it is a failed IP address System does not connect to server	System connection unsuccessful	Ensure invalid IP addresses do not enter the server

			with provided IP address		
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Disconnect:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
Successful Disconnection	The Client and Server are both running concurrently.	User selects disconnect	User selects disconnect from server Clients is successfully disconnected	User successfully disconnects from the server	Ensure the user is able to disconnect from the server
Failure to Disconnect	The Client and Server are both running concurrently.	User selects disconnect but client fails to disconnect from server	User selects disconnect from server Client fails to disconnect from server Client is still connected to the server System sends a message	User unsuccessfully disconnects from server	Check that the system sends a message when client is not disconnected
Ransom Disconnection	The Client and Server are both running concurrently.	Client loses connection	Client loses wifi connection Client disconnects from the server	Sudden user disconnection from server	Check when there is sudden connection loss, the client automatically disconnects

Password Recovery:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
Existing Email	User inputs wrong password 3 times and is locked out of the server then presses password reset button	User inputs existing username	User inputs username - JayBalz Client checks if username is in the system and is tied to an email Success - user gets one time code	Username is linked to an existing email	Ensure that the username is linked to an email
Non-Existent Email	User inputs wrong password 3 times and is locked out of the server then presses password reset button	User inputs a non-existent username	User inputs username - JayBallz Client checks if username is in the system and is tied to an email Fail - user does not get an email	Username is not linked to an email	Ensure that the invalid username does not connect to an email
Server Error	User inputs wrong password 3 times and is locked out of the server then presses password reset button	User inputs existing username but server times outs	User inputs username - JayBalz Client checks if username is in the system and tied to an email System and client fail to respond	System fails to respond, server timeout	System terminates when system fails to respond

Logout:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
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Manual Logout	Client connected and authenticated	User selects logout	User clicks logout. Session ends. System returns to login screen.	User session ends properly.	To confirm manual logout functionality.
Auto Logout	Client connected and authenticated	User is inactive for a while or shutdown	System detects inactivity Auto-logout is triggered. Session ends. System returns to login screen.	User is automatically logged out.	To check what happens when a user is inactive.

New User Registration:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
New Account (Valid)	Client and Server Running, User has no account.	Username: Nico Password: ImAWESOME Email: nico@gmail.com	User provides valid inputs. System verifies and stores the info.	New user is created	To validate successful new user registration.
New Account (Invalid)	Client and Server Running, User has no account.	Username: Nico Password: ImSAD (invalid), then ImAWESOME (valid) Email: nico@gmail.com	User provides invalid inputs. System rejects. User provides valid inputs. System verifies and stores the info.	User retries and successfully creates an account.	To check validation and retry behavior.

Shutdown:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
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Shutdown while logged out	System running, user logged out	User requests shutdown	User clicks shutdown. System terminates the client.	System terminates properly.	To ensure safe shutdown.
Shutdown while logged in	System running, user logged in	User requests shutdown	User clicks shutdown. System logs out user. System terminates the client.	User is logged out and the system terminates.	To ensure session safety during shutdown.
Unexpected Shutdown while Logged In	System running, user logged in	Computer dies/crashes	Unexpected shutdown. System logs out users automatically.	Session terminates correctly after crash	To confirm crash safety
Shutdown from Server GUI	System running, user logged in	Admin closes GUI	Admin closes server GUI. System terminates server.	Server is safely closed.	To validate server shutdown behavior.

Display Client/Server Info:

Test Name	Preconditions	Inputs	Procedures	Oracle	Rationale
Display Info	Server is running	Admin clicks query button	Admin clicks to view info. System fetches and displays stats.	Admin sees registered/logged-in /locked-out user info	To confirm correct display of server data

